

# Exogenous Reservoir Volatility with an Exponential-Quadratic Readout

Current specification, physical stylised facts, and joint equity–volatility–index calibration architecture

**Purpose.** This document consolidates the current exogenous-reservoir candidate. It states the model under the physical and pricing measures, identifies the mechanisms supporting each targeted stylised fact, and separates mathematical implications from numerical evidence and open calibration questions.

**Status:** research specification, not a calibration result

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## Abstract

The objective is a continuous-path stochastic-volatility model that can reproduce, under the physical measure, finite-frequency heavy tails, volatility clustering, roughness, leverage, a strong Zumbach effect and, at lower priority, the Taylor effect. Under the pricing measure, the same structural model should be capable of fitting Standard & Poor's 500 (S&P 500, whose option market is denoted SPX) and Cboe Volatility Index (VIX) option surfaces. The computational constraint is that changes in readout and risk-premium parameters must not require new simulation of the exogenous state paths.

The proposed model uses a structured bank of Ornstein–Uhlenbeck (OU) factors as a linear reservoir, an optional bounded nonlinear echo-state layer, and a normalised exponential-quadratic variance readout. A shifted quadratic projection of a return-correlated but variance-independent OU bank supplies leverage and time-reversal asymmetry. Slow exogenous OU modes supply clustering; a rough-kernel lift supplies finite-scale roughness; a fast, partly return-correlated direction supplies VIX-like spikes; and an independent quadratic block is reserved for dispersion or tail tuning. A deterministic Girsanov kernel preserves Gaussianity of the OU core between the physical probability measure  $\mathbb{P}$  and the risk-neutral pricing measure  $\mathbb{Q}$ .

The architecture supports all requested mechanisms without a quintic polynomial. This is a capacity statement, not a proof that a single parameter vector fits every physical statistic and both option surfaces simultaneously. The main unresolved questions are empirical joint calibration, identifiability, and the placement of the upper tempering level. With bounded variance, true stock martingality follows on finite horizons by Novikov; without tempering, the exponential-quadratic moment and martingale domains must be enforced explicitly.

## Contents

|          |                                                                   |          |
|----------|-------------------------------------------------------------------|----------|
| <b>1</b> | <b>Objective and standard of claim</b>                            | <b>1</b> |
| <b>2</b> | <b>The current model</b>                                          | <b>1</b> |
| 2.1      | Driver system and measure convention . . . . .                    | 1        |
| 2.2      | Layer 1: structured OU reservoir . . . . .                        | 2        |
| 2.3      | Layer 2: optional bounded echo-state layer . . . . .              | 2        |
| 2.4      | Exponential-quadratic readout . . . . .                           | 3        |
| 2.5      | Equity dynamics under $\mathbb{P}$ and $\mathbb{Q}$ . . . . .     | 4        |
| 2.6      | Architecture at a glance . . . . .                                | 5        |
| <b>3</b> | <b>Parameter hierarchy and path reuse</b>                         | <b>5</b> |
| 3.1      | Hyperparameters, structural parameters and daily inputs . . . . . | 5        |
| 3.2      | What path reuse does and does not mean . . . . .                  | 6        |
| <b>4</b> | <b>Physical-measure stylised facts</b>                            | <b>7</b> |
| 4.1      | Negligible linear autocorrelation of returns . . . . .            | 7        |
| 4.2      | Volatility clustering . . . . .                                   | 7        |
| 4.3      | Finite-scale roughness . . . . .                                  | 7        |
| 4.4      | Leverage effect . . . . .                                         | 8        |

|          |                                                                        |           |
|----------|------------------------------------------------------------------------|-----------|
| 4.5      | Strong Zumbach effect . . . . .                                        | 9         |
| 4.6      | Heavy-tailed returns at relevant frequencies . . . . .                 | 9         |
| 4.7      | Taylor effect . . . . .                                                | 10        |
| 4.8      | Fast volatility, downside asymmetry and VIX spike relaxation . . . . . | 11        |
| 4.9      | Summary of physical-measure support . . . . .                          | 11        |
| <b>5</b> | <b>Risk-neutral pricing and joint SPX–VIX calibration</b>              | <b>12</b> |
| 5.1      | Forward variance, VIX and VIX futures . . . . .                        | 12        |
| 5.2      | Analytic Gaussian-core moments . . . . .                               | 12        |
| 5.3      | Tempering and the nonlinear ESN layer . . . . .                        | 13        |
| 5.4      | SPX pricing and martingality . . . . .                                 | 13        |
| 5.5      | Why a quintic polynomial is not structurally required . . . . .        | 14        |
| <b>6</b> | <b>Joint calibration programme</b>                                     | <b>14</b> |
| 6.1      | Recommended nested sequence . . . . .                                  | 14        |
| 6.2      | Relation of M2 to ESN–A2 . . . . .                                     | 14        |
| 6.3      | Calibration hierarchy . . . . .                                        | 18        |
| 6.4      | Validation and falsification tests . . . . .                           | 18        |
| <b>7</b> | <b>Open issues and limitations</b>                                     | <b>19</b> |
| <b>8</b> | <b>Conclusion</b>                                                      | <b>20</b> |
| <b>A</b> | <b>Derivation of the Gaussian quadratic transform</b>                  | <b>20</b> |
| <b>B</b> | <b>Derivation of the Taylor benchmark</b>                              | <b>21</b> |
| <b>C</b> | <b>Related model classes</b>                                           | <b>21</b> |

## 1 Objective and standard of claim

Let  $S$  denote the total-return-adjusted equity index level and  $V$  its instantaneous variance. Throughout,  $\mathbb{P}$  denotes the physical probability measure governing historical dynamics, while  $\mathbb{Q}$  denotes the risk-neutral probability measure used for derivative pricing. The programme has two linked objectives:

1. Under  $\mathbb{P}$ , reproduce the relevant empirical properties of equity returns and volatility at intraday and daily frequencies.
2. Under  $\mathbb{Q}$ , reproduce the cross section of SPX options, VIX futures and VIX options, while reusing the same exogenous random paths during calibration.

The phrases *supports* and *reproduces* must not be conflated. Four evidence levels are used throughout:

| Label | Meaning                      | Interpretation                                                                           |
|-------|------------------------------|------------------------------------------------------------------------------------------|
| C     | By construction              | Follows directly from the state or readout definition.                                   |
| P     | Conditional proposition      | Follows under explicit non-degeneracy, moment or scale assumptions.                      |
| N     | Numerical mechanism evidence | Demonstrated by an ablation or controlled simulation, but not yet by market calibration. |
| O     | Open empirical question      | Must be established on historical data or option surfaces.                               |

### Main conclusion

There is no known structural incompatibility between the requested physical-measure facts, path reuse, and joint SPX–VIX pricing in the proposed class. There is also no theorem that a successful joint fit must exist. The correct conclusion is architectural feasibility subject to explicit moment, martingale and calibration tests.

## 2 The current model

### 2.1 Driver system and measure convention

Work on a filtered probability space satisfying the usual conditions. Under a measure  $\mathbb{M} \in \{\mathbb{P}, \mathbb{Q}\}$ , let  $W^{S,\mathbb{M}}$  be the standard Brownian motion driving the equity return. The superscript  $S$  labels the equity, and  $\mathbb{M}$  identifies the probability measure. The structured reservoir uses four driver groups:

$$dW_t^{F,\mathbb{M}} = \rho_F dW_t^{S,\mathbb{M}} + \sqrt{1 - \rho_F^2} dB_t^{F,\mathbb{M}}, \quad (1)$$

$$dW_t^{J,\mathbb{M}} = \rho_J dW_t^{S,\mathbb{M}} + \sqrt{1 - \rho_J^2} dB_t^{J,\mathbb{M}}, \quad (2)$$

where  $B^{F,\mathbb{M}}$  and  $B^{J,\mathbb{M}}$  are independent standard Brownian motions, also independent of  $W^{S,\mathbb{M}}$ . The labels  $F$  and  $J$  denote the feedback and fast-spike banks, while  $\rho_F$  and  $\rho_J$  are their instantaneous correlations with the equity Brownian motion. Consequently  $W^{F,\mathbb{M}}$  and  $W^{J,\mathbb{M}}$

are standard Brownian motions correlated with the equity driver. The clustering drivers  $B^{C,\mathbb{M}}$  and optional orthogonal drivers  $B^{O,\mathbb{M}}$  are further independent standard Brownian motions, mutually independent and independent of  $W^S, B^F, B^J$ . Partial rather than zero correlation is recommended for the fast bank, because large VIX moves are empirically associated with equity sell-offs.

## 2.2 Layer 1: structured OU reservoir

For each bank  $b \in \{F, J, C, O\}$ , define under  $\mathbb{P}$  the Ornstein–Uhlenbeck (OU) coordinates

$$dX_{j,t}^{b,\mathbb{P}} = -\lambda_j^b X_{j,t}^{b,\mathbb{P}} dt + \sqrt{2\lambda_j^b} dW_{j,t}^{b,\mathbb{P}}, \quad j = 1, \dots, N_b. \quad (3)$$

Here  $X_{j,t}^{b,\mathbb{P}}$  is coordinate  $j$  of bank  $b$  at time  $t$ ,  $\lambda_j^b > 0$  is its mean-reversion rate,  $W_j^{b,\mathbb{P}}$  is the Brownian motion assigned to that coordinate, and  $N_b$  is the number of coordinates in the bank. The diffusion coefficient  $\sqrt{2\lambda_j^b}$  normalises every stationary marginal variance to one.

Within the feedback bank, the coordinates share  $W^F$ ; this creates the correlated multi-rate lift used to approximate a rough Volterra kernel. The fast bank can share one or a small number of partially return-correlated drivers. The clustering and orthogonal banks may use independent drivers, so that their covariance is a flexible positive mixture of exponentials.

For  $N_b \geq 2$ , the rate grids are geometric and fixed in the inner calibration:

$$\lambda_j^b = \lambda_{\min}^b \left( \frac{\lambda_{\max}^b}{\lambda_{\min}^b} \right)^{\frac{j-1}{N_b-1}}. \quad (4)$$

Thus  $\lambda_{\min}^b$  and  $\lambda_{\max}^b$  are respectively the slowest and fastest mean-reversion rates in bank  $b$ .

Define normalised projections

$$Y_t = a^\top X_t^F, \quad F_t = f^\top X_t^J, \quad \text{Var}_{\mathbb{P}}(Y_t) = \text{Var}_{\mathbb{P}}(F_t) = 1 \quad (5)$$

at the stationary physical reference point. Here  $X_t^F$  and  $X_t^J$  are the vectors of feedback- and fast-bank coordinates, while the fixed vectors  $a$  and  $f$  aggregate each bank into one scalar factor. The normalisation gives both projections unit stationary variance, so their readout coefficients have comparable interpretations.

The projection  $Y$  is a persistent, multiscale summary of past return-correlated shocks. Its shifted quadratic term below produces the main leverage effect—negative returns followed by higher volatility—and the Zumbach time-reversal effect. The projection  $F$  emphasises the fastest return-correlated modes. Its role is to create the sharp front of a volatility spike and its subsequent mean-reverting decay. Shifting the quadratic function of  $F$  makes a negative equity move generate a larger spike than a positive move of equal magnitude.

## 2.3 Layer 2: optional bounded echo-state layer

An echo-state network (ESN) is an optional second reservoir layer with fixed recurrent weights and a calibrated output loading. On a time grid  $t_n = n\Delta t$ , where  $n$  is the time-step index,  $\Delta t > 0$  is the step length and  $0 < \alpha \leq 1$  is the leak parameter, define

$$Z_{n+1} = (1 - \alpha)Z_n + \alpha \tanh(A_Z Z_n + B_Z \psi(X_n) + b_Z), \quad (6)$$

where  $Z_n$  is the second-layer state,  $X_n$  is the complete first-layer OU state,  $A_Z$  is the recurrent matrix,  $B_Z$  is the input matrix,  $b_Z$  is a bias vector, and  $\psi$  is a fixed feature map that may contain selected linear, quadratic and cross-bank features. The matrices and bias are drawn or designed once and then held fixed. A sufficient contraction condition for the echo-state property is

$$(1 - \alpha) + \alpha \|A_Z\|_2 < 1. \quad (7)$$

The echo-state property is important because it gives the recurrent layer fading memory: for the same input history, the effect of two different initial states decays geometrically. The state is therefore a stable causal function of the exogenous OU history rather than an additional source of persistent initial-condition dependence. This prevents exploding or ambiguous recurrent dynamics and makes simulations, calibration results and path reuse reproducible after a finite washout period.

Because  $Z$  is driven only by the exogenous OU state, neither  $X$  nor  $Z$  changes when a readout coefficient changes. The layer is called optional precisely because it affects variance only through the term  $\delta^\top Z_t$  in the readout below. When the loading vector  $\delta = 0$ , that term vanishes and Layer 2 can be omitted without changing  $V$  or  $S$ . When  $\delta \neq 0$ , bounded nonlinear combinations of the OU history enter log variance and the full two-layer ESN is active. The recommended first calibration uses  $\delta = 0$  and activates the layer only if an out-of-sample ablation demonstrates an incremental gain large enough to justify the extra parameters and the loss of Gaussian conditional-moment formulas.

#### Terminology

With  $\delta = 0$ , the operative model is a structured Gaussian reservoir rather than a nonlinear ESN in the narrow, classical sense. It remains an ESN-style fixed-reservoir model. The full two-layer ESN is obtained when  $\delta^\top Z_t$  enters the readout.

## 2.4 Exponential-quadratic readout

The multiplier-free log-variance score is

$$\eta_t = c_C^\top X_t^C + q_Z(Y_t - \beta)^2 + q_J(F_t - \beta_J)^2 + \frac{q_O}{N_O} \|X_t^O\|^2 + \delta^\top Z_t. \quad (8)$$

The score  $\eta_t$  is dimensionless. The vector  $X_t^C$  contains the slow clustering coordinates and  $c_C$  is their linear loading. The non-negative curvatures  $q_Z$ ,  $q_J$  and  $q_O$  control respectively the feedback, fast-spike and optional orthogonal quadratic channels. The constants  $\beta$  and  $\beta_J$  shift the vertices of the feedback and fast parabolas. The vector  $X_t^O$  contains  $N_O$  orthogonal coordinates, so division by  $N_O$  keeps their average squared energy comparable across bank sizes. Finally,  $\delta$  is the loading vector of the optional echo-state layer  $Z_t$ .

The recommended core imposes  $q_Z > 0$ ,  $q_J \geq 0$ ,  $\beta_J \geq 0$ , and starts at  $q_O = 0$ ,  $\delta = 0$ . Setting  $\beta_J = 0$  centres the fast parabola at the stationary state  $F = 0$ . Its conversion of positive and negative fast-factor movements of equal magnitude into volatility spikes is then symmetric. A positive  $\beta_J$ , together with positive fast-factor/equity correlation, breaks this symmetry in the empirically relevant direction: negative index movements produce larger volatility spikes. Each shifted quadratic is deliberately parameterised without an additional linear loading on the same projection: adding both  $\ell F$  and  $q_J(F - \beta_J)^2$ , for example, would make the linear loading and shift redundant up to deterministic normalisation.

For a cap level  $L$  and sharpness  $\kappa > 0$ , define the smooth upper tempering function

$$h_{L,\kappa}(x) = L - \frac{1}{\kappa} \log\left(1 + e^{\kappa(L-x)}\right), \quad G_{L,\kappa}(x) = e^{2h_{L,\kappa}(x)}. \quad (9)$$

It satisfies  $h_{L,\kappa}(x) \approx x$  well below  $L$  and  $h_{L,\kappa}(x) \uparrow L$  as  $x \rightarrow \infty$ . Let  $\xi_0(t)$  be the market forward variance curve and let  $\underline{v}$  be a small deterministic variance floor. The variance is

$$V_t = \underline{v} + (\xi_0(t) - \underline{v}) \frac{G_{L,\kappa}(\eta_t)}{M_{\mathbb{Q}}(t)}, \quad M_{\mathbb{Q}}(t) = \mathbb{E}_0^{\mathbb{Q}}[G_{L,\kappa}(\eta_t)]. \quad (10)$$

Here  $L$  is the upper level approached by the tempered score,  $\kappa$  controls how sharply the transition occurs,  $G_{L,\kappa}$  maps the score into a positive variance multiplier, and  $M_{\mathbb{Q}}(t)$  is its time- $t$  risk-neutral expectation conditional on time-zero information. The division by  $M_{\mathbb{Q}}(t)$  preserves the prescribed forward variance curve.

Provided  $\xi_0(t) > \underline{v}$ , positivity and  $\mathbb{E}_0^{\mathbb{Q}}[V_t] = \xi_0(t)$  follow by construction. The untempered shadow model is obtained by setting  $h(x) = x$ ; it is useful for analytic checks and control variates, but it is not the recommended production specification on a return-correlated quadratic bank.

Tempering is introduced because an exponential of a Gaussian quadratic can assign enormous variance to rare factor realisations. Without tempering, admissibility depends on spectral moment conditions: some parameter values have infinite variance moments, Monte Carlo estimates can be dominated by a handful of paths, numerical optimisation becomes unstable, and the stock martingale condition is no longer automatic. The upper bound supplied by  $h_{L,\kappa}$  ensures finite variance moments of every order and gives a direct finite-horizon Novikov bound, while leaving ordinary observations almost unchanged when  $\eta_t \ll L$ . What is sacrificed is exact Gaussian exponential-quadratic transform formulae and completely unbounded spike amplitudes. The untempered shadow model is therefore retained for analytic diagnostics, whereas the tempered model is used for robust simulation and calibration.

## 2.5 Equity dynamics under $\mathbb{P}$ and $\mathbb{Q}$

Let  $r_t$  be the instantaneous risk-free rate and  $d_t$  the continuous dividend yield. Under the physical measure, use a deterministic instantaneous Sharpe ratio  $\theta(t)$ :

$$\frac{dS_t}{S_t} = (r_t - d_t + \theta(t)\sqrt{V_t}) dt + \sqrt{V_t} dW_t^{S,\mathbb{P}}. \quad (11)$$

The name Sharpe ratio is literal at the instantaneous level: the expected equity return in excess of  $r_t - d_t$  is  $\theta(t)\sqrt{V_t}$ , while its instantaneous volatility is  $\sqrt{V_t}$ . Their ratio is therefore  $\theta(t)$ . This specification lets the physical equity premium vary with volatility without feeding the variance level back into the exogenous reservoir dynamics.

Collect the independent primitive Brownian motions into a vector  $\mathbf{W}$ , whose entries are  $W^S, B^F, B^J, B^C$  and  $B^O$ , with as many components in the last two groups as their banks require. Let this vector satisfy the deterministic Girsanov shift

$$d\mathbf{W}_t^{\mathbb{P}} = d\mathbf{W}_t^{\mathbb{Q}} - \gamma(t) dt, \quad \gamma_S(t) = \theta(t). \quad (12)$$

The vector  $\gamma(t)$  contains the market prices of risk of the independent Brownian sources. Its equity component  $\gamma_S$  must equal  $\theta$  so that the physical equity premium disappears under  $\mathbb{Q}$ .

The other components need not equal  $\theta$ : they price shocks that are not fully spanned by the equity return and determine how the conditional means of the volatility factors differ between  $\mathbb{P}$  and  $\mathbb{Q}$ . Thus  $\theta$  is both the instantaneous equity Sharpe ratio and one component of the broader Brownian risk-premium vector; the remaining  $\gamma$ 's refer to different primitive shock equations, not to alternative definitions of the equity Sharpe ratio.

Then

$$\frac{dS_t}{S_t} = (r_t - d_t) dt + \sqrt{V_t} dW_t^{S,\mathbb{Q}}. \quad (13)$$

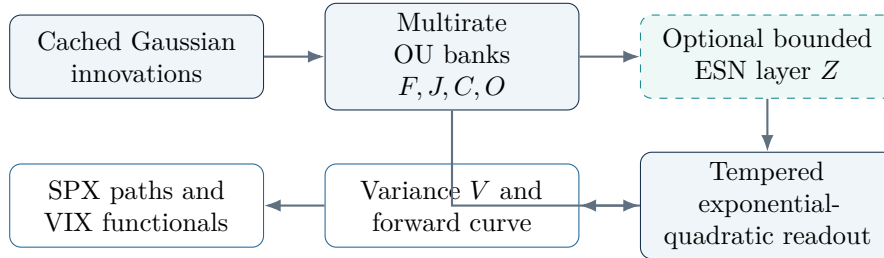
The unspanned components of  $\gamma$  are parsimonious variance-risk-premium parameters. Because  $\gamma$  is deterministic, every OU bank remains Gaussian under  $\mathbb{Q}$ : only its deterministic conditional mean changes, while its covariance and rate spectrum are preserved. This is the essential reason the same centred Gaussian innovations can be reused under both measures.

More explicitly, if  $\gamma_b(t)$  denotes the effective deterministic risk premium of the Brownian driver assigned to coordinate  $(b, j)$ , then

$$dX_{j,t}^{b,\mathbb{Q}} = \left( -\lambda_j^b X_{j,t}^{b,\mathbb{Q}} - \sqrt{2\lambda_j^b} \gamma_b(t) \right) dt + \sqrt{2\lambda_j^b} dW_{j,t}^{b,\mathbb{Q}}. \quad (14)$$

For example, the feedback driver in the first equation is a linear combination of two primitive Brownian motions, so its effective premium is  $\gamma_F = \rho_F \theta + \sqrt{1 - \rho_F^2} \gamma_{BF}$ ; the fast-bank formula is analogous. This is why the coordinate-level  $\gamma_b$ 's can differ even though the equity component is fixed by  $\gamma_S = \theta$ .

## 2.6 Architecture at a glance



## 3 Parameter hierarchy and path reuse

### 3.1 Hyperparameters, structural parameters and daily inputs

| Class                         | Examples                                                                                                        | Treatment                                                                                                                                       |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| Architectural hyperparameters | Bank dimensions; OU rate grids; reservoir matrices $A_Z, B_Z$ ; leak $\alpha$ ; feature map $\psi$ ; cap family | Selected in an outer validation loop and changed infrequently. Changing OU rates or fixed reservoir weights requires rebuilding state features. |

| Class                        | Examples                                                     | Treatment                                                                                                                                                                                                          |
|------------------------------|--------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Shared structural parameters | $a, f, c_C, q_Z, \beta, q_J, q_O, \delta, \rho_F, \rho_J$    | Estimated from a panel of physical and option data with regularisation. Readout changes do not change cached states; correlation changes require rebuilding correlated states from the same primitive innovations. |
| Measure-dependent parameters | $\theta$ and unspanned components of $\gamma$                | Thin deterministic parameterisation. They shift conditional means but do not change OU covariances.                                                                                                                |
| Daily inputs and states      | $\xi_0(\cdot)$ , rates, dividends, spot, filtered $X_t, Z_t$ | Updated daily. They are inputs or filtered states, not structural calibration parameters.                                                                                                                          |

### 3.2 What path reuse does and does not mean

For fixed hyperparameters and correlations, simulate or reconstruct once:

$$\{\Delta W^S, \Delta B^F, \Delta B^J, \Delta B^C, \Delta B^O; X; Z\}. \quad (15)$$

When  $q_Z, \beta, q_J, q_O, c_C, \delta$  or the deterministic risk premia change, no new random numbers are required. The readout, variance, stock integral and payoffs must nevertheless be recomputed. It is impossible for the final  $S$  or VIX paths to remain unchanged when the volatility readout changes.

| Change                           | New random draw?                                | Required recomputation                                              |
|----------------------------------|-------------------------------------------------|---------------------------------------------------------------------|
| Readout coefficient              | No                                              | $\eta, V, S$ , VIX map and payoffs                                  |
| Deterministic risk premium       | No                                              | Deterministic OU mean shift; possibly $Z$ ; then pricing quantities |
| Forward-variance curve           | No                                              | Variance level, $S$ , VIX and payoffs                               |
| Correlation                      | No, if primitive shocks are cached              | Correlated drivers and affected OU states                           |
| OU rate grid or reservoir matrix | No new seed is necessary, but the state changes | Full feature-engine reconstruction                                  |

This is the same computational principle exploited by precomputed signature and quantisation approaches: expensive stochastic input is generated offline, while parameter-dependent maps are evaluated during optimisation [4, 2].

## 4 Physical-measure stylised facts

The selected targets are standard empirical regularities of equity markets, with roughness and time-reversal asymmetry adding more discriminating requirements to the classical inventory [3, 5, 8].

### 4.1 Negligible linear autocorrelation of returns

For non-overlapping increments over a short interval  $\Delta$ , the martingale part of equation (11) has conditional mean zero and variance of order  $\Delta$ . The predictable return component is of order  $\Delta$ , so its contribution to return covariance is of order  $\Delta^2$ . Consequently, linear return autocorrelation is small at intraday and daily frequencies unless the risk-premium term is made too large or too state dependent.

Under  $\mathbb{Q}$ , discounted simple gains are martingale increments and therefore orthogonal across non-overlapping intervals. Log returns need not be exactly uncorrelated because their drift contains  $-V_t/2$ .

**Status: P/O.** Near-zero autocorrelation is structurally natural, not an identity under  $\mathbb{P}$ . It must remain an explicit diagnostic, especially after estimating  $\theta$ .

### 4.2 Volatility clustering

The recommended architecture obtains clustering from the exogenous slow bank rather than the endogenous activity filter. If  $C_t = c_C^\top X_t^C$  and the coordinates are independent stationary OU processes, then

$$\text{Cov}(C_t, C_{t+\tau}) = \sum_{j=1}^{N_C} c_{C,j}^2 e^{-\lambda_j^C \tau}. \quad (16)$$

A geometrically spaced rate grid approximates slowly decaying covariance over a prescribed finite interval. The monotone exponential link transfers persistence from  $\eta$  to  $V$ ; quadratic channels add modes with decay rates such as  $2\lambda_j$  and  $\lambda_i + \lambda_j$ .

This choice preserves strong frozen-state reuse. An endogenous activity bank

$$d\Sigma_{k,t} = \alpha_k (V_t - \Sigma_{k,t}) dt \quad (17)$$

remains an optional challenger if the exogenous bank cannot fit the empirical volatility ACF. It needs no new random numbers, but it must be recomputed whenever the readout changes.

**Status: C/P/O.** Persistence is built into the rate mixture; matching its empirical magnitude and shape is a calibration question.

### 4.3 Finite-scale roughness

Rough volatility is usually expressed through the small-lag scaling

$$\mathbb{E}[(\log V_{t+\tau} - \log V_t)^2] \asymp C\tau^{2H}, \quad H < \frac{1}{2}, \quad (18)$$

with empirical estimates often near  $H = 0.1$  [5]. A fractional kernel can be approximated on a chosen range by a positive exponential sum,

$$K_H(t) \approx \sum_{j=1}^{N_F} a_j e^{-\lambda_j^F t}, \quad (19)$$

which is precisely a Markovian OU lift [1]. The feedback projection  $Y = a^\top X^F$  therefore reproduces equation (18) over the frequencies covered by the rate grid.

If  $g$  is the smooth map from the state to log variance, then

$$g(Y_{t+\tau}) - g(Y_t) = g'(Y_t)(Y_{t+\tau} - Y_t) + o_{\mathbb{P}}(\tau^H). \quad (20)$$

Thus the exponent is locally preserved whenever  $g'(Y_t) \neq 0$ . The derivative can vanish at the parabola vertex and becomes small near cap saturation, so roughness should be estimated on the bulk of the volatility distribution.

#### Finite lift, finite claim

Every finite OU system is a semimartingale and has Brownian  $H = 1/2$  behaviour at sufficiently small asymptotic lags. The defensible claim is rough behaviour on the intraday-to-daily scale range selected by  $[\lambda_{\min}^F, \lambda_{\max}^F]$ , not genuine fractional asymptotics at arbitrarily small times.

**Status: P/O.** The lift supports the target scaling over a finite band; the realised band and estimated  $H$  must be verified with a noise-robust realised-volatility estimator.

#### 4.4 Leverage effect

For a lag  $\tau > 0$ , define the leverage statistic

$$L(\tau) = \text{Corr}_{\mathbb{P}}(r_t, V_{t+\tau}), \quad L(\tau) < 0 \text{ empirically.} \quad (21)$$

The feedback contribution is  $q_Z(Y - \beta)^2$ , with derivative

$$\frac{\partial \eta}{\partial Y} = 2q_Z(Y - \beta). \quad (22)$$

Near the stationary centre  $Y \approx 0$ , its sign is  $-\text{sign}(q_Z\beta)$ . Since the outer link is increasing away from cap saturation, the instantaneous return-log-variance covariation has leading sign

$$\text{sign}(d\langle \log S, \log V \rangle_t) \approx -\text{sign}(q_Z\beta\rho_F). \quad (23)$$

Under the present convention,  $q_Z > 0$ ,  $\beta > 0$ ,  $\rho_F > 0$  therefore produce negative leverage. This apparently unusual positive correlation is a consequence of the shifted parabola convention; reversing both  $\beta$  and  $\rho_F$  leaves the local sign unchanged.

The shift is essential. With  $\beta = 0$ , the quadratic channel is locally even around the stationary centre and does not generate a stable directional leverage effect. The fast bank should not dominate total variance, because a large orthogonal or symmetric component dilutes correlation-based leverage statistics.

**Status: P/N/O.** The sign mechanism is explicit and has been observed in the programme's simulations. Its empirical magnitude remains a calibration constraint.

#### 4.5 Strong Zumbach effect

The Zumbach effect is a causal time-reversal asymmetry between past returns and future volatility. A representative statistic at horizon  $\tau$  is

$$Z(\tau) = \text{Corr}(R_{t-\tau,t}^2, \text{RV}_{t,t+\tau}) - \text{Corr}(\text{RV}_{t-\tau,t}, R_{t,t+\tau}^2). \quad (24)$$

A reversible exogenous volatility process independent of returns has  $Z(\tau) = 0$  in the ideal stationary limit. The current model breaks reversibility without making the OU driver amplitude depend on  $V$ :

1. the feedback state is a causal filter of a shock correlated with the return innovation;
2. the future variance depends quadratically on that state;
3. past return shocks therefore affect future variance through  $Y^2$ , while the time-reversed pairing does not have the same causal construction.

The programme's controlled Gate C experiment provides mechanism evidence:

| Configuration                                     | Pearson asymmetry |
|---------------------------------------------------|-------------------|
| GARCH positive control                            | $0.118 \pm 0.009$ |
| Standardised correlated driver, quadratic readout | $0.297 \pm 0.011$ |
| Quadratic-ablation reference run                  | $0.215 \pm 0.018$ |
| Same ablation run with $q_Z = 0$                  | $0.000 \pm 0.002$ |

These are replicated mechanism checks, not market calibration results. They show that the quadratic readout is load-bearing and that variance-dependent raw-return driving is not required. The standardised-driver sweep and the quadratic ablation used different reference runs, so their absolute levels should not be compared as if they were one nested sample. This differs from endogenous price-feedback models such as quadratic rough Heston, but it targets the same time-reversal asymmetry [8, 6].

##### Measurement requirement

In the programme's validation, replacing realised variance by daily squared returns removed almost all power of the Zumbach statistic, even on a GARCH positive control. The empirical target should therefore use intraday realised variance, and every result should be accompanied by a reversible null, a positive control and replication uncertainty.

**Status: N/O.** The mechanism is strongly supported by ablation. Its market magnitude cannot be claimed before a correctly measured intraday-data study.

#### 4.6 Heavy-tailed returns at relevant frequencies

Conditional on the variance path, short-horizon returns are approximately Gaussian; their unconditional distribution is a variance mixture. The exponential-quadratic channel produces large finite-frequency dispersion. In the scalar untempered illustration

$$\sigma = e^{qY^2}, \quad Y \sim \mathcal{N}(0, 1), \quad q > 0, \quad (25)$$

one has, up to a slowly varying logarithmic factor,

$$\mathbb{P}(\sigma > x) = \mathbb{P}\left(Y^2 > \frac{\log x}{q}\right) \asymp x^{-1/(2q)}(\log x)^{-1/2}. \quad (26)$$

This explains why an exponential quadratic can generate much heavier observable tails than a direct quadratic variance map. It does not, by itself, establish the tail index of a continuous-time return when the volatility state and return share Brownian increments.

For  $X \sim \mathcal{N}(m, \Sigma)$  and a quadratic matrix  $Q$ , the untempered first variance moment requires

$$I - 4\Sigma^{1/2}Q\Sigma^{1/2} \succ 0. \quad (27)$$

Higher variance moments impose progressively stronger spectral conditions. The tempered link removes asymptotic moment explosion and replaces true power tails by a broad but ultimately truncated mixture. This is consistent with the stated objective: realistic heavy-tail behaviour over observable intraday and daily quantiles is more important than an exact asymptotic law.

The optional orthogonal block can increase log-volatility dispersion without changing the largest quadratic eigenvalue when its loading is spread over many independent directions and each new eigenvalue remains below the existing maximum. It should be activated only if the correlated feedback and fast banks cannot match return kurtosis and volatility quantiles without damaging leverage or Zumbach.

**Status: P/N/O.** Heavy finite-frequency tails are a strong capability of the readout. Their quantitative fit and stability across aggregation horizons remain empirical tests.

#### 4.7 Taylor effect

The Taylor effect requires

$$\text{Corr}(|r_t|, |r_{t+k}|) > \text{Corr}(r_t^2, r_{t+k}^2) \quad (28)$$

over a relevant set of lags. It is not implied by leverage, roughness or clustering alone. For the benchmark representation  $r_t = e^{X_t}\varepsilon_t$ , where  $X$  is stationary Gaussian with variance  $s^2$ , lag correlation  $\rho_k$ , and  $\varepsilon_t \stackrel{\text{iid}}{\sim} \mathcal{N}(0, 1)$ , direct calculation gives

$$\rho_{|r|}(k) = \frac{(2/\pi) \left( e^{s^2\rho_k} - 1 \right)}{e^{s^2} - 2/\pi}, \quad (29)$$

$$\rho_{r^2}(k) = \frac{e^{4s^2\rho_k} - 1}{3e^{4s^2} - 1}. \quad (30)$$

At small  $s^2$ , the first correlation grows more slowly, so the Taylor inequality fails. At  $\rho_k = 0.9$ , the crossover in this benchmark is approximately  $s^2 = 0.081$ , or  $\text{sd}(\log \sigma) \approx 0.285$ . The threshold changes with persistence and with departures from the Gaussian log-volatility benchmark.

The exponential-quadratic and optional multi-directional blocks provide the required dispersion more naturally than a direct quadratic variance readout. Nevertheless, Taylor must remain a diagnostic or low-weight target rather than a theorem of the architecture.

**Status: P/N/O.** The model can cross the analytically identified dispersion threshold; the sign and lag range must be checked after every joint calibration.

#### 4.8 Fast volatility, downside asymmetry and VIX spike relaxation

Suppose a fast OU projection is shocked to  $F_t = f_0$ . Ignoring subsequent noise,

$$F_{t+s} \approx f_0 e^{-\lambda_J s}. \quad (31)$$

Its untempered contribution to relative variance is

$$\frac{V_{t+s}}{\bar{V}_{t+s}} \approx \exp\left(2q_J \left[f_0^2 e^{-2\lambda_J s} - 2\beta_J f_0 e^{-\lambda_J s}\right]\right), \quad (32)$$

where  $\bar{V}$  uses the resting fast state  $F = 0$ . For equal-magnitude shocks  $f > 0$ ,

$$\log \frac{V(F = -f)}{V(F = +f)} = 8q_J \beta_J f. \quad (33)$$

Thus  $q_J > 0$ ,  $\beta_J > 0$ , and positive fast-bank/equity correlation make a negative equity shock produce a larger contemporaneous volatility response than an equal positive shock. The result is exact below the tempering region; the smooth cap can attenuate it in the far tail. The zero-shift restriction is the symmetric nested control.

The corresponding time derivative is

$$\frac{d}{ds} \log \left( \frac{V_{t+s}}{\bar{V}_{t+s}} \right) = -4\lambda_J q_J \left[ f_0^2 e^{-2\lambda_J s} - \beta_J f_0 e^{-\lambda_J s} \right]. \quad (34)$$

For the downside shocks of interest ( $f_0 < 0$ ), this derivative is negative and its magnitude is largest near the event before slowing progressively. Several fast and medium OU rates create a sharp front followed by a persistent shoulder. This temporal shape comes from the interaction of mean reversion with the exponential level map, not from a jump in the state process.

##### Fresh-seed mechanism screen

A paired screen of 16 previously selected configuration–scenario pairs, five fresh seeds and six values  $\beta_J \in \{0, 0.1, \dots, 0.5\}$  produced 480 evaluations. At  $\beta_J = 0.1$ , the downside-minus-upside median day-zero volatility-response gap increased from 35.0% to 38.7%, and the corresponding days 1–5 gap increased from 29.4% to 37.3%. All seven pre-existing stylised-fact proxy gates passed in 67 of 80 evaluations; six of 16 configurations passed on every fresh seed. All failures at this shift came from the positive-Zumbach lower bound. This is controlled synthetic mechanism evidence, not an empirical SPX calibration.

The cap may flatten the very top of an extreme spike. Its level should be selected so that it almost never binds in the calibration region while still controlling martingality outside it.

**Status: P/N/O.** The relaxation shape follows from the model and the downside orientation has controlled numerical support. Matching the distribution, amplitude and equity co-movement of observed VIX spikes is an empirical requirement.

#### 4.9 Summary of physical-measure support

| Target                          | Status | Load-bearing component                                                     |
|---------------------------------|--------|----------------------------------------------------------------------------|
| Near-zero return ACF            | P/O    | Diffusive return innovation and parsimonious deterministic Sharpe ratio    |
| Volatility clustering           | C/P/O  | Slow exogenous OU mixture                                                  |
| Finite-scale roughness          | P/O    | Shared-driver rough OU lift and appropriate rate grid                      |
| Leverage                        | P/N/O  | Shift $\beta$ , curvature $q_Z$ , and return correlation $\rho_F$          |
| Strong Zumbach                  | N/O    | Causal correlated OU filter plus quadratic readout                         |
| Heavy finite-frequency tails    | P/N/O  | Exponential quadratic; optional orthogonal dispersion block                |
| Taylor effect                   | P/N/O  | Sufficient log-volatility dispersion and persistence                       |
| Fast asymmetric VIX-like spikes | P/N/O  | Shifted fast, partly correlated OU projection in the exponential quadratic |

## 5 Risk-neutral pricing and joint SPX–VIX calibration

### 5.1 Forward variance, VIX and VIX futures

By equation (10), the input curve satisfies

$$\mathbb{E}_0^{\mathbb{Q}}[V_t] = \xi_0(t). \quad (35)$$

For a VIX window  $\Delta = 30/365$ , ignoring the conventional discrete-sampling and corridor adjustments,

$$\text{VIX}_T^2 = \frac{100^2}{\Delta} \int_T^{T+\Delta} \mathbb{E}_T^{\mathbb{Q}}[V_u] du. \quad (36)$$

Matching  $\xi_0$  does not automatically match VIX futures, because

$$F_0^{\text{VIX}}(T) = \mathbb{E}_0^{\mathbb{Q}}[\text{VIX}_T] \leq \sqrt{\mathbb{E}_0^{\mathbb{Q}}[\text{VIX}_T^2]}. \quad (37)$$

The difference is the VIX convexity adjustment and must be calibrated or reported, not hidden inside the forward-variance normalisation.

### 5.2 Analytic Gaussian-core moments

If  $\delta = 0$  and  $h(x) = x$ , the OU state is conditionally Gaussian and the score is quadratic. If

$$X_u \mid \mathcal{F}_T \sim \mathcal{N}(m_{T,u}, \Sigma_{T,u}) \quad (38)$$

and the exponent is  $X^\top A X + b^\top X + c$ , then

$$\mathbb{E}_T \left[ e^{X^\top A X + b^\top X + c} \right] \quad (39)$$

$$= \frac{\exp \left( c + m^\top A m + b^\top m + \frac{1}{2} (2A m + b)^\top (I - 2\Sigma A)^{-1} \Sigma (2A m + b) \right)}{\sqrt{\det(I - 2\Sigma A)}}, \quad (40)$$

provided

$$I - 2\Sigma^{1/2} A \Sigma^{1/2} \succ 0. \quad (41)$$

This turns equation (36) into a one-dimensional deterministic maturity quadrature evaluated at the cached state  $X_T$ . No nested Monte Carlo is needed in the Gaussian shadow model.

### 5.3 Tempering and the nonlinear ESN layer

The recommended tempered model does not retain the closed form equation (40). Neither does the full model with  $\delta \neq 0$ . It remains computationally tractable through one of the following offline–online decompositions:

1. low-dimensional Gauss–Hermite or sparse-grid quadrature when the active quadratic directions are few;
2. functional quantisation of the Gaussian state;
3. least-squares conditional expectation on cached states;
4. a neural surrogate trained on the state and parameter domain, with the Gaussian core used as a control variate.

Quantisation and neural pricing have already been demonstrated for Gaussian polynomial SPX–VIX models [2]; precomputed signature samples provide another successful offline calibration paradigm [4]. These are useful numerical templates, not proofs that the present model will calibrate.

### 5.4 SPX pricing and martingality

For fixed cached innovations and a candidate parameter vector, compute

$$\log \frac{S_T}{S_0} = \int_0^T \left( r_t - d_t - \frac{1}{2} V_t \right) dt + \int_0^T \sqrt{V_t} dW_t^{S, \mathbb{Q}}. \quad (42)$$

The discounted stock is always a nonnegative local martingale. It is a true martingale if the stochastic exponential has expectation one. In the tempered model,  $V$  is bounded on every finite horizon when  $\xi_0$  and the deterministic risk premia are bounded and the positive normaliser  $M_{\mathbb{Q}}$  is bounded away from zero. Novikov’s condition then gives

$$\mathbb{E}^{\mathbb{Q}} \left[ \exp \left( \frac{1}{2} \int_0^T V_t dt \right) \right] < \infty, \quad (43)$$

so the discounted stock is a true martingale. In the untempered, return-correlated exponential-quadratic model, finite marginal moments do not by themselves establish true martingality; the admissible domain must be analysed separately.

**Production choice**

The smooth upper tempering is part of the recommended model, not merely a numerical emergency stop. It provides a clean finite-horizon martingale argument while preserving the untempered map over the empirically relevant state region.

**5.5 Why a quintic polynomial is not structurally required**

The fifth-degree polynomial used by Abi Jaber, Illand and Li is effective in their one-factor Gaussian polynomial family and helps generate upward VIX smiles [2]. It is not a necessity theorem. Two complementary results make this clear:

- quadratic rough Heston combines a quadratic variance map with rough feedback and provides a continuous-path solution to the joint SPX–VIX smile problem [6];
- signature models obtain strong joint calibrations using several low-order path features rather than a scalar quintic readout [4].

The present architecture substitutes feature diversity, rate diversity and exponential convexity for scalar polynomial degree. Its fast and slow quadratic directions have distinct economic roles, which should improve identifiability relative to an unrestricted high-degree polynomial.

**6 Joint calibration programme****6.1 Recommended nested sequence**

| Model | Readout                                                 | Purpose                                                                                                     |
|-------|---------------------------------------------------------|-------------------------------------------------------------------------------------------------------------|
| M0    | Positive direct quadratic variance                      | Minimal baseline; tests whether exponential convexity is genuinely needed.                                  |
| M1    | Untempered Gaussian exponential quadratic, $\delta = 0$ | Analytic shadow model; moment-domain and VIX-control-variate benchmark.                                     |
| M2    | Tempered Gaussian exponential quadratic, $\delta = 0$   | Recommended production core.                                                                                |
| M3    | M2 plus $\delta^\top Z$                                 | Activate only if the nonlinear ESN layer materially improves out-of-sample physical facts or option errors. |
| M4    | M3 plus orthogonal block $q_O > 0$                      | Optional finite-frequency dispersion and tail tuning; activate only after diagnosing a residual deficiency. |

If M2 cannot reproduce synthetic surfaces generated by a quintic OU benchmark, add a second identified quadratic direction before increasing polynomial degree. This directly tests the hypothesis that factor diversity can replace degree.

**6.2 Relation of M2 to ESN–A2**

M2 is a structured descendant of ESN–A2, but it is neither a renaming nor a nested restriction of A2. The common core is a fixed multirate Gaussian reservoir whose state can be reused while the volatility readout changes. The redesign concerns the organisation of that state, the readout parameterisation, and the separation between physical and pricing inputs. The comparison

below refers to the supplied reconstructed A2 engine; its implementation file records that it was reconstructed after an environment reset and validated against the reported roughness targets. Consequently, code-specific A2 conclusions carry that provenance.

On a grid of width  $\Delta$ , the A2 rough bank and nonlinear layer can be written schematically as

$$r_{n+1} = D_\lambda r_n + c_\lambda \varepsilon_{n+1}, \quad D_{\lambda,jj} = e^{-\lambda_j \Delta}, \quad c_{\lambda,j} = \sqrt{1 - e^{-2\lambda_j \Delta}}, \quad (44)$$

$$z_{n+1} = \Phi_{A2}(r_{n+1}, z_n), \quad (45)$$

where the same scalar innovation  $\varepsilon_{n+1}$  drives every coordinate of  $r$  and the equity return in the supplied simulator. The map  $\Phi_{A2}$  is the leaky, bounded second layer built from quadratic and linear features, a hyperbolic tangent, local recurrence, and the state-dependent norm gate. Its volatility score and variance map are

$$\eta_n^{A2} = b_0 + \kappa q^\top r_n + w_z^\top z_n + \frac{m_1}{N_r} \|r_n\|^2 + \frac{m_2}{N_r} (q^\top r_n)^2, \quad (46)$$

$$V_n^{A2} = \sigma_{\min}^2 + \text{softplus}(\eta_n^{A2})^2. \quad (47)$$

At the A2 resting bias, softplus is already close to its exponential branch. Thus A2 already contains the central idea “Gaussian reservoir plus approximately exponential-quadratic readout.” In contrast, M2 sets  $\delta = q_O = 0$  in equation (8) and uses

$$\eta_t^{M2} = c_C^\top X_t^C + q_Z(Y_t - \beta)^2 + q_J(F_t - \beta_J)^2, \quad (48)$$

$$V_t^{M2} = \underline{v} + (\xi_0(t) - \underline{v}) \frac{G_{L,\kappa}(\eta_t^{M2})}{M_Q(t)}. \quad (49)$$

The models therefore share a mathematical backbone, but they are not the same parameterised family:

| Component            | ESN–A2                                                                                                                                                 | M2                                                                                                                                                                        |
|----------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| State drivers        | One multirate rough bank driven by the equity innovation; the coupling is effectively perfect in the supplied engine.                                  | Separate feedback, fast-spike, and clustering banks. Feedback and spike shocks may be only partially correlated with the equity shock; clustering shocks are independent. |
| Nonlinear layer      | The bounded recurrent $z$ -layer contributes directly through $w_z^\top z$ .                                                                           | Inactive: $\delta = 0$ . M3, not M2, restores an optional fixed contractive echo layer.                                                                                   |
| Leverage and Zumbach | The linear rough projection, aligned quadratic projection, and nonlinear layer have overlapping effects, so attribution is parameter-regime dependent. | Concentrated in the shifted correlated channel $q_Z(Y - \beta)^2$ , whose slope controls the leverage sign and whose state dependence generates time-reversal asymmetry.  |
| Clustering           | Encoded jointly by the multirate bank, especially its isotropic reservoir-energy term, and persistence transmitted through the nonlinear layer.        | A dedicated slow linear bank $c_C^\top X^C$ , separated from the leverage–Zumbach channel.                                                                                |

| Component              | ESN–A2                                                                                                                                         | M2                                                                                                                                                                                                |
|------------------------|------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Fast VIX-like spikes   | No separately identified fast-spike bank.                                                                                                      | A dedicated convex direction $q_J(F - \beta_J)^2$ , with its own fast rate grid and return correlation; $\beta_J > 0$ orients equal-magnitude responses toward downside shocks.                   |
| Variance map and tails | Squared softplus with an unbounded exponential branch; the quadratic eigenvalues govern its asymptotic moment domain.                          | A direct exponential-quadratic shadow map, smoothly tempered in production and close to the shadow map below the cap. The target is finite-frequency fat tails, not an asymptotic power law.      |
| Measure change         | The supplied engine is a physical-measure simulator and does not give a complete, separately normalised $\mathbb{P}/\mathbb{Q}$ specification. | Explicit $\mathbb{P}$ and $\mathbb{Q}$ stock equations, deterministic Girsanov kernel, unspanned risk premia, and a market forward-variance normalisation.                                        |
| Path reuse             | Available while parameters that determine $r$ and $z$ are fixed; changing reservoir or second-layer parameters rebuilds the state.             | Primitive shocks and exogenous states are cached. Readout, forward-curve, and deterministic risk-premium changes require deterministic state adjustments and revaluation, but no new random draw. |
| Calibration roles      | The audited parameter split allowed quantities intended for daily recalibration to move most of the physical statistics.                       | Architecture, shared structural parameters, measure-dependent premia, and daily inputs are separated explicitly; shared readout coefficients cannot be overwritten by a daily $\mathbb{Q}$ fit.   |

The shifted M2 channel makes the relationship particularly transparent:

$$q_Z(Y - \beta)^2 = q_Z Y^2 - 2q_Z \beta Y + q_Z \beta^2. \quad (50)$$

It combines A2’s aligned linear and quadratic rough-factor channels, but imposes the identified coefficient relation  $\kappa_Y = -2q_Z \beta$  after normalisation. Conversely, baseline M2 drops A2’s isotropic energy and nonlinear readouts, replaces their clustering role by  $X^C$ , and adds the fast direction  $F$ . M4 can restore an independent quadratic dispersion channel, but that channel lives in an orthogonal bank and is therefore not equivalent to A2’s  $\|r\|^2$ . M3 restores nonlinear memory, but its controlled echo layer does not make the resulting model identical to A2 either.

#### Nesting statement

A2 is not a parameter point of M2, and M2 is not a parameter point of A2. M2 is best viewed as a lower-nonlinearity but more economically structured redesign of A2. It retains the exogenous Gaussian and exponential-quadratic mechanisms, while changing the driver topology, assigning separate channels to clustering and spikes, enforcing path-reuse discipline, and adding the objects required for joint SPX–VIX pricing.

The transfer of stylised-fact evidence is therefore conditional. A2 experiments motivate the

rough-reservoir and convex-readout mechanisms, but they do not prove that M2 preserves their empirical magnitudes. M2 must be evaluated on its own with matched estimators, multiple seeds, and the ablations prescribed below. Likewise, M2 has better specified joint-pricing capacity, not a demonstrated SPX–VIX fit in the absence of option data.

### 6.3 Calibration hierarchy

Let  $\Theta_H$ ,  $\Theta_S$ ,  $\Theta_M$  and  $D_t$  denote architectural hyperparameters, shared structural parameters, measure-dependent parameters and daily inputs. A practical hierarchy is:

1. **Architecture selection.** Choose dimensions and rate grids using rolling physical-measure validation and synthetic option-capacity tests. This is the expensive outer loop.
2. **Shared structural estimation.** Estimate  $\Theta_S$  from historical windows and a panel of option surfaces, with stability penalties across dates.
3. **Measure change.** Estimate the thin deterministic unspanned risk premia in  $\Theta_M$ , with  $\theta$  pinned by the physical drift convention.
4. **Daily calibration.** Strip  $\xi_0$ , filter the current state and calibrate only the permitted daily pricing quantities.

Because shared readout parameters affect both  $\mathbb{P}$  and  $\mathbb{Q}$ , a purely sequential "fit  $\mathbb{P}$ , then overwrite under  $\mathbb{Q}$ " procedure is not valid. Use either a joint loss or alternating minimisation with a physical-measure retention penalty:

$$\mathcal{L}(\Theta) = w_P \mathcal{L}_P + w_S \mathcal{L}_{\text{SPX}} + w_V \mathcal{L}_{\text{VIX}} + w_F \mathcal{L}_{\text{VIX futures}} + \mathcal{R}(\Theta). \quad (51)$$

Option residuals should be expressed in bid–ask-normalised prices, for example

$$\varepsilon_i = \frac{C_i^{\text{model}} - C_i^{\text{mid}}}{\max\{\frac{1}{2}(C_i^{\text{ask}} - C_i^{\text{bid}}), \epsilon_i^{\text{floor}}\}}, \quad (52)$$

rather than unweighted implied-volatility errors. VIX futures must be included separately from the forward-variance curve.

### 6.4 Validation and falsification tests

| Question  | Required test                                                                                 | Failure implication                                                  |
|-----------|-----------------------------------------------------------------------------------------------|----------------------------------------------------------------------|
| Roughness | Structure functions on noise-robust intraday realised volatility across the designed lag band | Retune rate grid; do not add readout degree.                         |
| Leverage  | Lagged return–realised-volatility correlation with confidence bands                           | Adjust $\beta$ , $\rho_F$ , or add a separate linear leverage bank.  |
| Zumbach   | Realised-variance statistic, reversible null, positive control, multiple seeds or blocks      | Reassess the correlated quadratic mechanism or measurement protocol. |

| Question               | Required test                                                             | Failure implication                                                         |
|------------------------|---------------------------------------------------------------------------|-----------------------------------------------------------------------------|
| Heavy tails            | QQ/body fit, kurtosis, threshold-stability plots and aggregation horizons | Add orthogonal dispersion only after excluding measurement and cap effects. |
| Taylor                 | Both correlation curves and the log-volatility dispersion threshold       | Treat as secondary unless failure contaminates more important targets.      |
| VIX spikes             | Peak amplitude, rise/decay asymmetry and SPX co-movement                  | Retune fast rates and $\rho_J$ ; avoid a dominant independent spike factor. |
| Martingale             | Analytic boundedness proof plus Monte Carlo defect within sampling error  | Reject parameter point or lower the cap.                                    |
| Joint surface capacity | Bid-ask-normalised SPX/VIX errors and VIX-futures errors over dates       | Add a second quadratic direction before quintic degree.                     |
| Path reuse             | Bitwise equality of cached $X, Z$ under readout perturbations             | Implementation violates the architectural constraint.                       |

## 7 Open issues and limitations

### What is not yet established

1. No SPX/VIX option data have yet been used, so joint calibration capacity is untested.
2. Identifiability of curvature, shift, correlations, fast rates and deterministic risk premia must be studied through Jacobian singular values, profile likelihoods and rolling stability.
3. The optimal cap level must balance negligible in-sample binding against robust moment and martingale control.
4. A finite OU lift produces only finite-scale roughness.
5. The exact market magnitude of the Zumbach effect requires intraday realised variance; daily squared returns are an inadequate proxy for the current statistic.
6. The nonlinear second layer is not currently justified by the Gate C Zumbach ablation. It must earn its parameters through out-of-sample improvement.

Three interactions deserve particular monitoring:

- Increasing an independent dispersion block can improve tails and Taylor while diluting leverage and Zumbach correlations.
- Increasing the correlated quadratic curvature strengthens Zumbach and tails but pushes the untempered model toward moment and martingale boundaries.
- A low cap improves safety but can flatten extreme VIX spikes, distort roughness estimates in the upper tail and limit the VIX smile.

These are not arguments against the model. They are the principal trade-offs that the joint objective must expose rather than hide.

## 8 Conclusion

The current candidate is a two-layer exogenous reservoir architecture with a recommended Gaussian first calibration and an optional nonlinear ESN extension. Its defining choices are:

- multirate OU banks with variance-independent drivers;
- a return-correlated rough feedback projection;
- a shifted quadratic feedback term for leverage and Zumbach;
- slow exogenous modes for clustering and path reuse;
- a shifted fast, partly return-correlated direction for downside-oriented VIX spike relaxation, nested at  $\beta_J = 0$ ;
- a normalised, smoothly tempered exponential-quadratic variance map;
- a deterministic market price of risk preserving the Gaussian OU class under  $\mathbb{Q}$ ;
- an optional independent dispersion block and optional nonlinear ESN layer, both subject to ablation.

This construction retains the mechanisms needed for the targeted physical-measure facts and offers a viable offline–online route to SPX/VIX calibration without a quintic polynomial. Roughness and path reuse are architectural; leverage has an explicit sign condition; Zumbach has controlled numerical support; tails and VIX spikes follow from exponential-quadratic convexity; and Taylor follows only above a measurable dispersion threshold.

The next decisive step is not another architectural elaboration. It is a disciplined implementation of M0–M2, validation on intraday physical data, and synthetic joint-calibration tests against quadratic-rough-Heston and quintic-OU benchmarks before real option data become available.

## A Derivation of the Gaussian quadratic transform

Let  $X = m + Z$ , with  $Z \sim \mathcal{N}(0, \Sigma)$ , and let  $A = A^\top$ . Write

$$X^\top AX + b^\top X + c = c + m^\top Am + b^\top m + u^\top Z + Z^\top AZ, \quad u = 2Am + b. \quad (53)$$

Whitening with  $Z = \Sigma^{1/2}U$  and completing the square in the Gaussian density gives

$$\mathbb{E}\left[e^{u^\top Z + Z^\top AZ}\right] = \det(I - 2\Sigma A)^{-1/2} \exp\left[\frac{1}{2}u^\top (I - 2\Sigma A)^{-1}\Sigma u\right], \quad (54)$$

provided  $I - 2\Sigma^{1/2}A\Sigma^{1/2} \succ 0$ . Substitution yields equation (40). For  $e^{2\eta}$  with quadratic matrix  $Q$ , set  $A = 2Q$ , which gives the condition in equation (27).

## B Derivation of the Taylor benchmark

Let  $r_t = e^{X_t} \varepsilon_t$ , where  $\mathbb{E}[X_t] = \mu$ ,  $\text{Var}(X_t) = s^2$ ,  $\text{Corr}(X_t, X_{t+k}) = \rho_k$ , and the standard-normal innovations are independent through time and independent of  $X$ . Then

$$\mathbb{E}[e^{X_t}] = e^{\mu+s^2/2}, \quad \mathbb{E}[e^{2X_t}] = e^{2\mu+2s^2}, \quad (55)$$

$$\mathbb{E}[e^{X_t+X_{t+k}}] = e^{2\mu+s^2(1+\rho_k)}, \quad \mathbb{E}[\varepsilon] = \sqrt{2/\pi}. \quad (56)$$

Using these moments in the covariance and variance of  $|r|$  yields the first formula in equation (30). Applying the same calculation to  $r^2$ , with  $\mathbb{E}[\varepsilon^4] = 3$ , gives the second. The result is a diagnostic approximation for the current nonlinear model, not an exact formula for it.

## C Related model classes

| Class                          | Main device                                                | Relation to the present model                                                                                                       |
|--------------------------------|------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Gaussian polynomial volatility | Polynomial of a Gaussian Volterra state                    | Demonstrates strong joint SPX–VIX calibration and motivates Gaussian path reuse; degree five is specific to that one-factor family. |
| Quadratic rough Heston         | Endogenous rough price feedback and quadratic variance     | Shows that a quadratic map can support strong Zumbach and joint smiles, but sacrifices exogenous frozen states.                     |
| Signature volatility           | Linear readout of rich path features                       | Demonstrates that feature diversity and offline samples can replace high scalar polynomial degree.                                  |
| Rough-volatility lift          | Exponential-sum approximation of fractional kernels        | Supplies the multirate Markovian mechanism used for finite-scale roughness.                                                         |
| Classical ESN                  | Fixed contractive recurrent reservoir plus trained readout | Supplies the optional nonlinear second layer and the path-reuse philosophy.                                                         |

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